Project Presentation

Housing Prices Prediction

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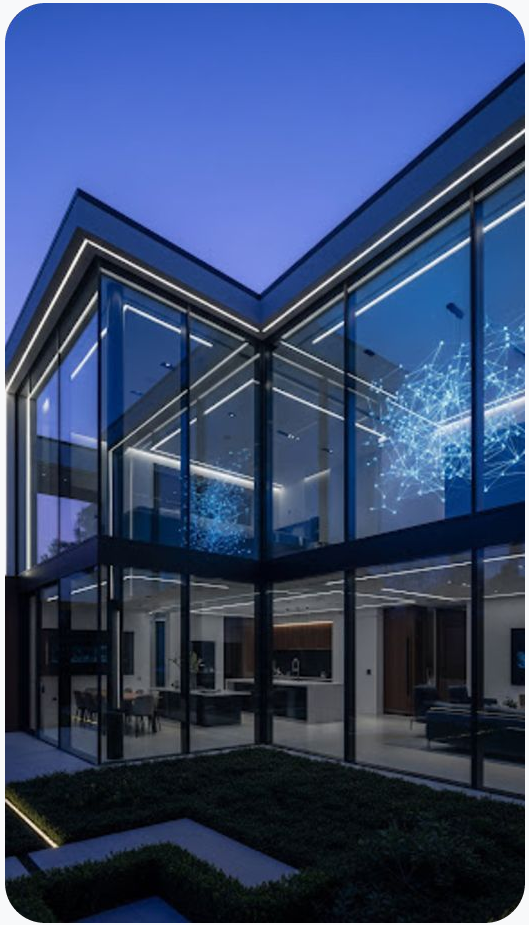


Strategic Analysis & Data Roadmap

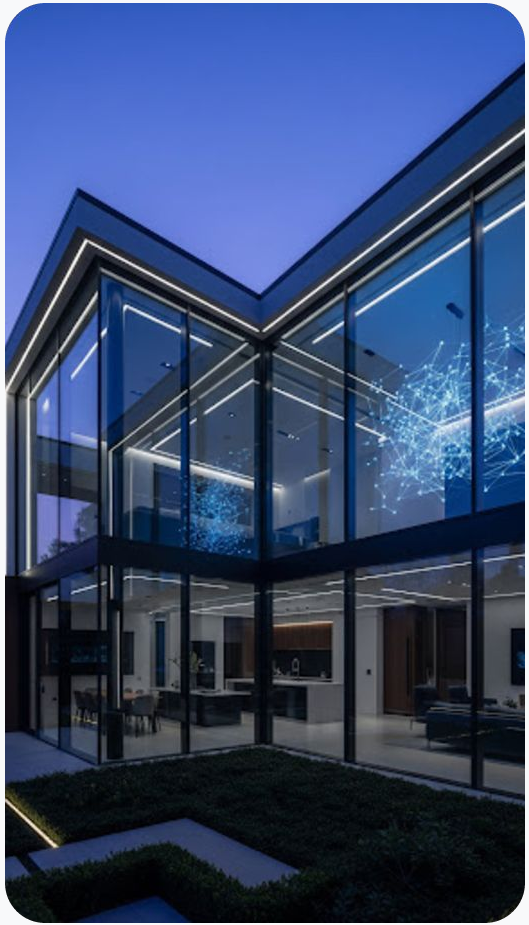
Problem Statement
Statement. Scope, causes and implications.

Data Overview
Input variables, target features, and validation sets

Methodology
EDA, ML models, and Least Squares fitting

Key Findings
Goodness of fit, residuals, and model accuracy

Business Recommendations
Strategic actions based on predictive demand and market trends



Strategic Analysis & Data Roadmap

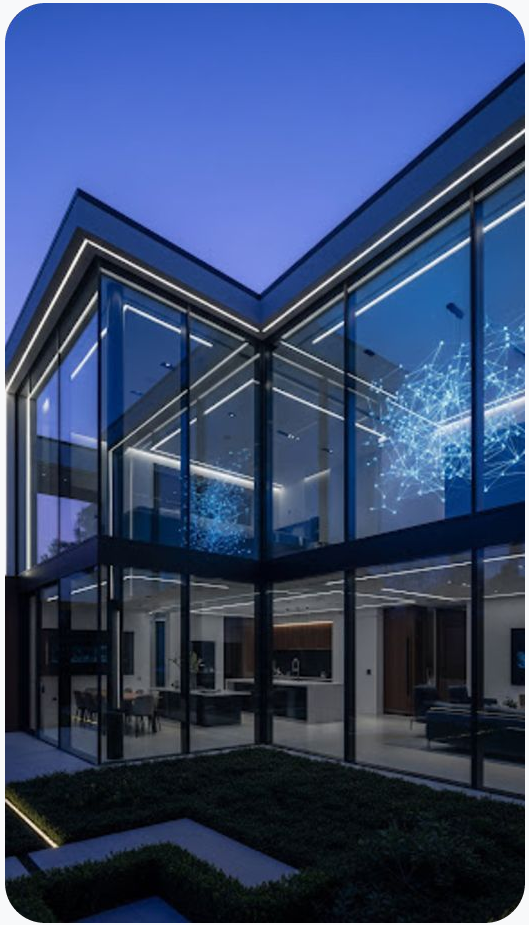
Problem Statement

Statement. Scope, causes and implications.

- **Problem Statement Definition.**

Motivation & Background

- **Metro Vancouver has one of the least affordable housing markets in Canada**
- **Housing prices are driven by location, land value, and economic activity**
- **Real-world analysis requires integrating multiple datasets**
- **We combine:**
  - **Property tax data**
  - **Business licences**
  - **Geographic boundaries**
  - **Crime**



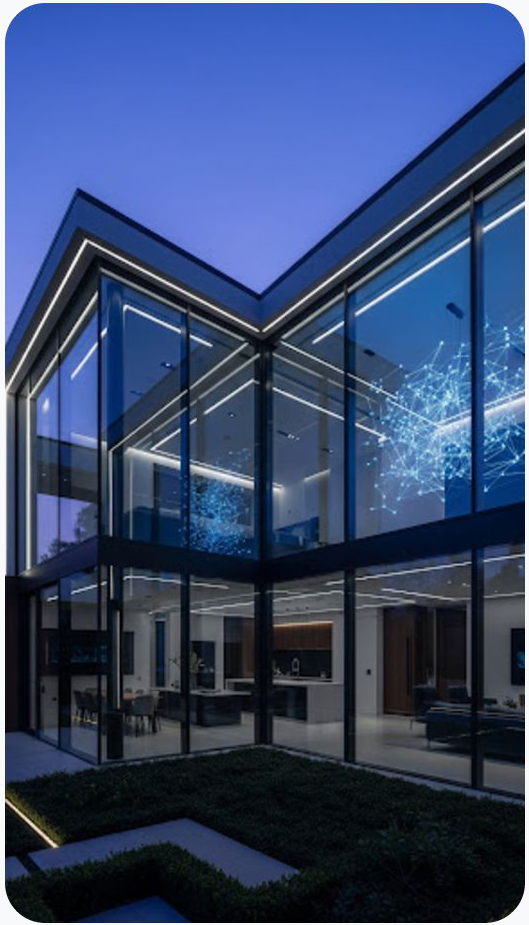
Strategic Analysis & Data Roadmap

Problem Statement

Statement. Scope, causes and implications.

- **What is the specific business problem you are trying to solve?**

“Analyze and predict housing value patterns in Metro Vancouver to identify key drivers and detect high-value or bubble - areas.”



Strategic Analysis & Data Roadmap

Problem Statement

Statement. Scope, causes and implications.

- **What are the scope, causes, and implications of the problem?**

Scope

- **Data: Property values + neighborhoods + business activity + Crime rate**
- **Analysis: EDA, feature engineering, predictive modeling**
- **Goal: *Understand price drivers across locations***

Causes

- **Supply vs demand imbalance**
- **Location premium (neighborhood effects)**
- **Business and economic activity concentration**
- **Limited land availability**

Implications

- **Housing affordability challenges**
- **Inequality across neighborhoods**
- **Risk of housing bubbles**
- **Need for better urban planning decisions**

Data Overview

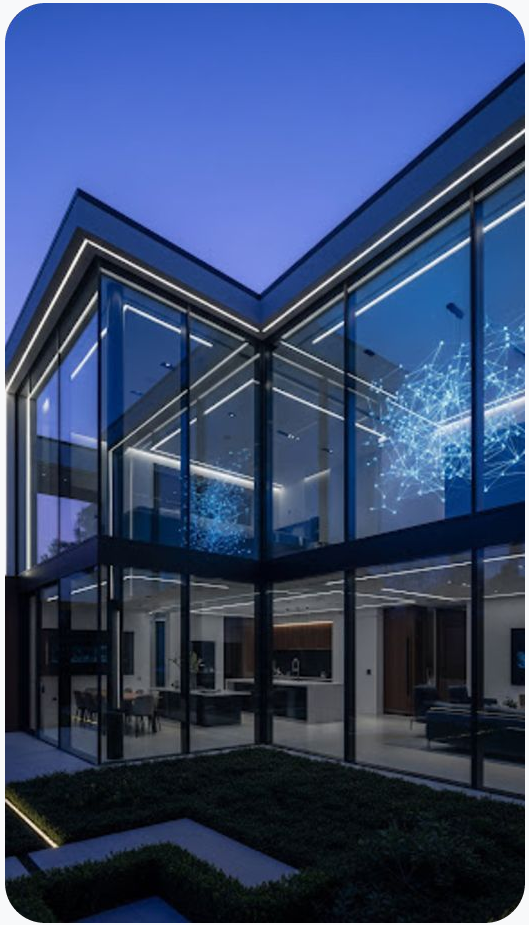
Input variables, target features, and validation sets

Datasets Used

- We used four integrated datasets to support house price prediction:
  a. Property Tax
  b. Business Licences
  c. Local Areas
  d. Crime
- The datasets include a mix of:
  a. Numerical variables for valuation and counts
  b. Categorical variables for neighborhood and type classifications
  c. Geographic information for spatial analysis
- The property tax dataset is the core source for the target variable, while the other datasets provide contextual predictors.

Dataset source:

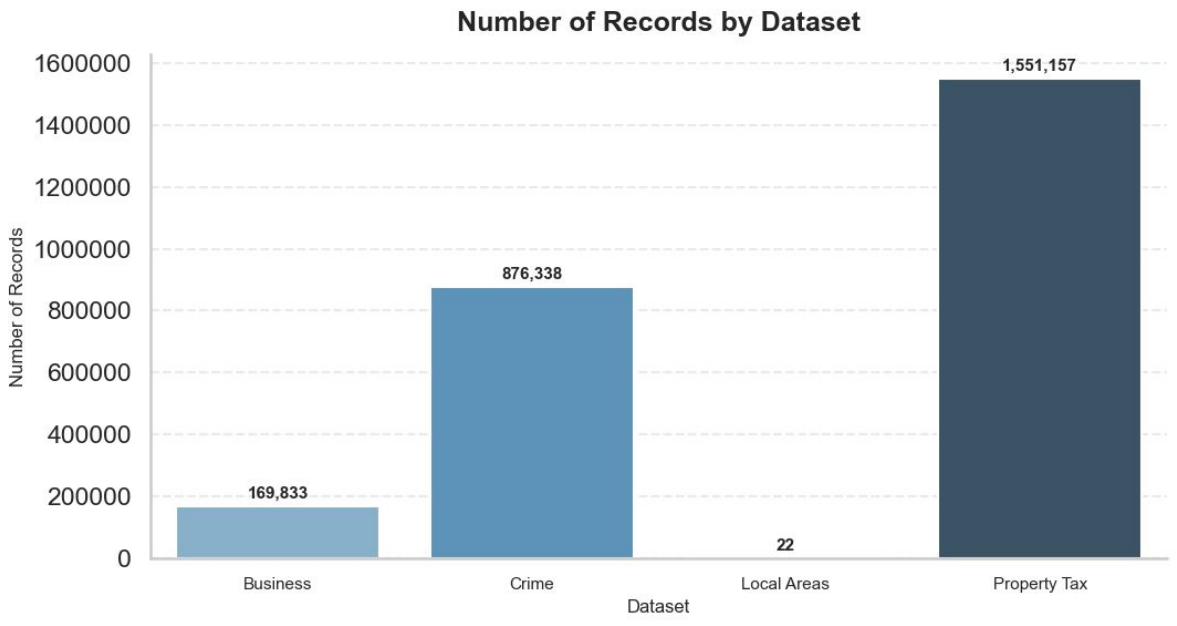
- Local Area Boundary: https://opendata.vancouver.ca/explore/dataset/local-area-boundary/
- Property Tax Report: https://opendata.vancouver.ca/explore/dataset/property-tax-report/
- Business Licences: https://opendata.vancouver.ca/explore/dataset/business-licences/
- Crime Statistics: https://geodash.vpd.ca/opendata/



Data Overview

Input variables, target features, and validation sets

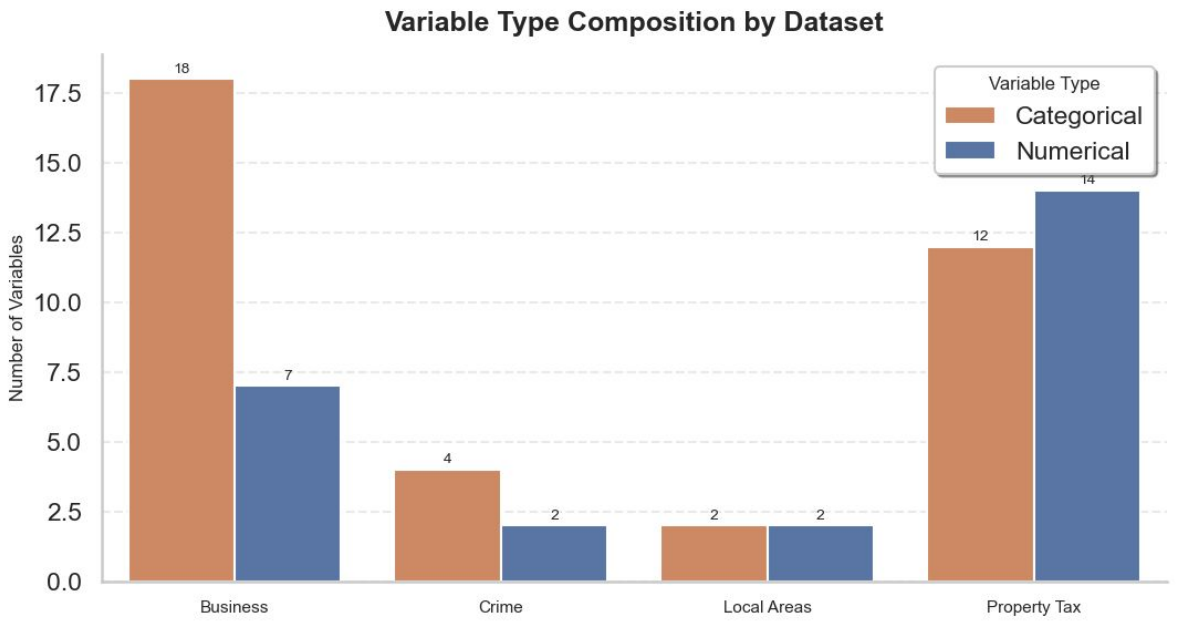
- **The property tax dataset is the largest dataset**
- **Crime data also contains a significant number of records**
- **Business data has a moderate size**
- **Local areas dataset is very small and used for reference**



Data Overview

Input variables, target features, and validation sets

- **Most datasets are composed of categorical variables**
- **The property tax dataset has the highest number of numerical variables**
- **Business data is mainly categorical, reflecting classification-based information**
- **The data structure supports both descriptive and predictive modeling**



Data Overview

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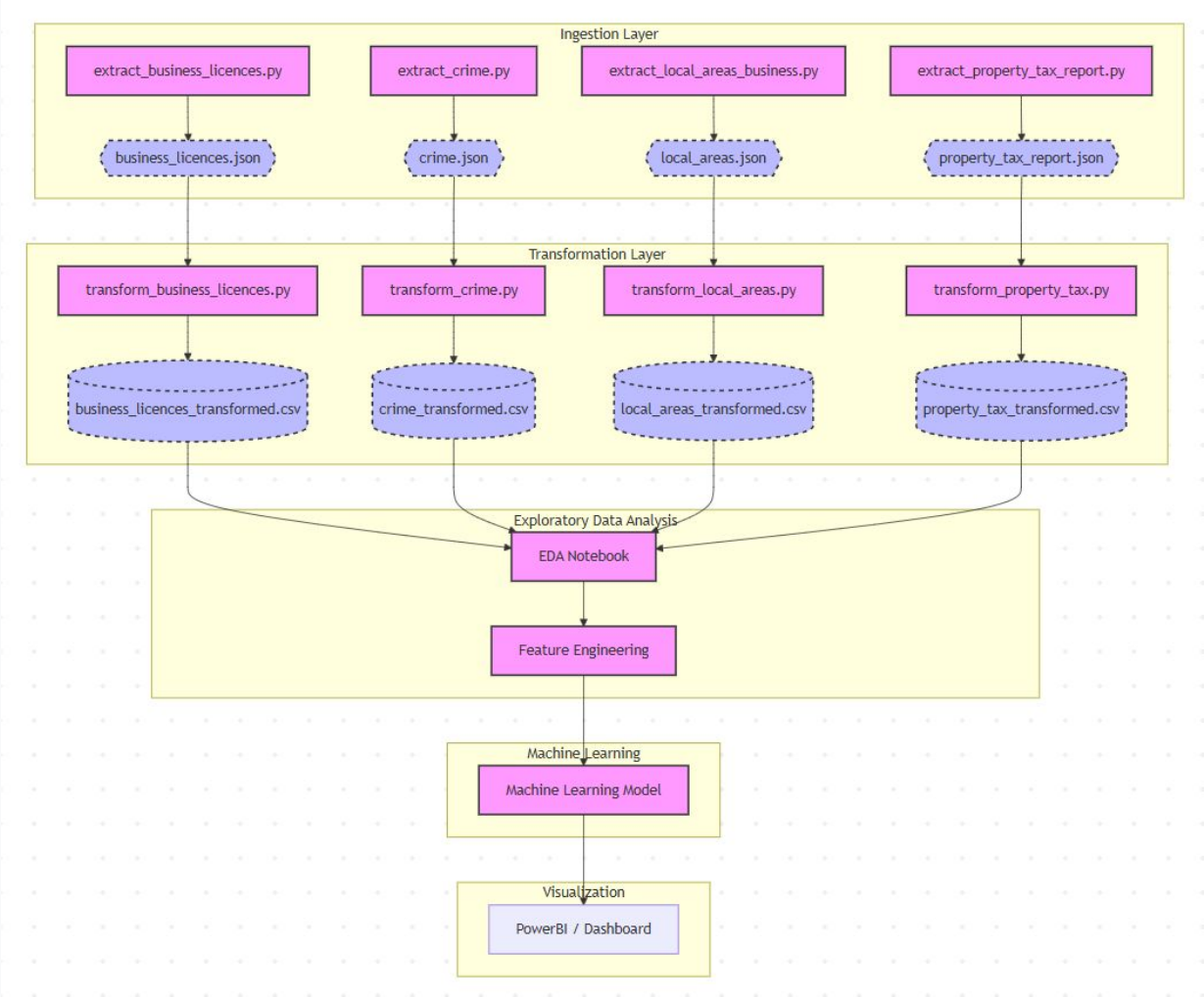
Data Preparation

- Data cleaning and formatting
- Merging datasets by geographic area
- Feature engineering:
- Total Value = Land + Improvement
- Business density per area

Key Variables for Modeling

Target: Total Assessed Value

Predictors: Land Value, Improvement Value, Local Area, Business Density, Crime Rate



Project Architecture

Directed Acyclic Graph

Strategic Analysis & Data Roadmap

Comprehensive overview from problem statement to business impact

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Defining core challenges and regression objectives

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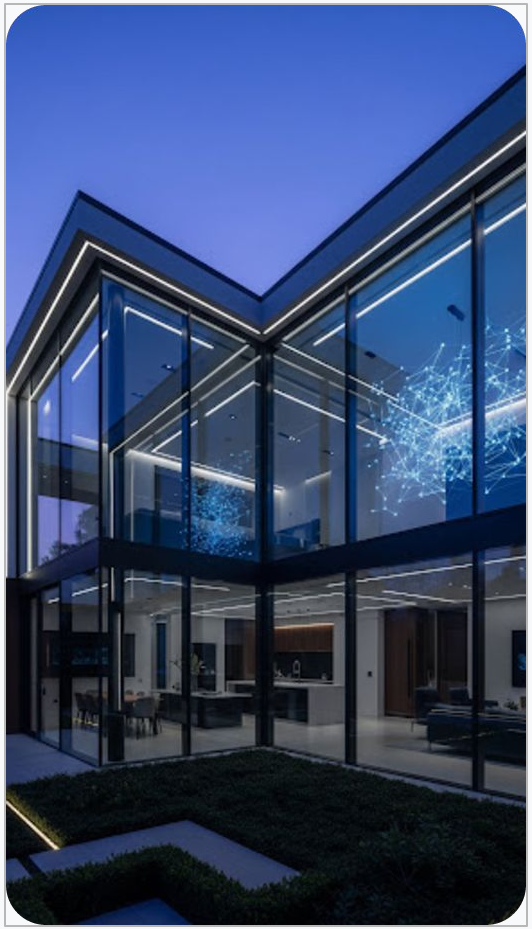


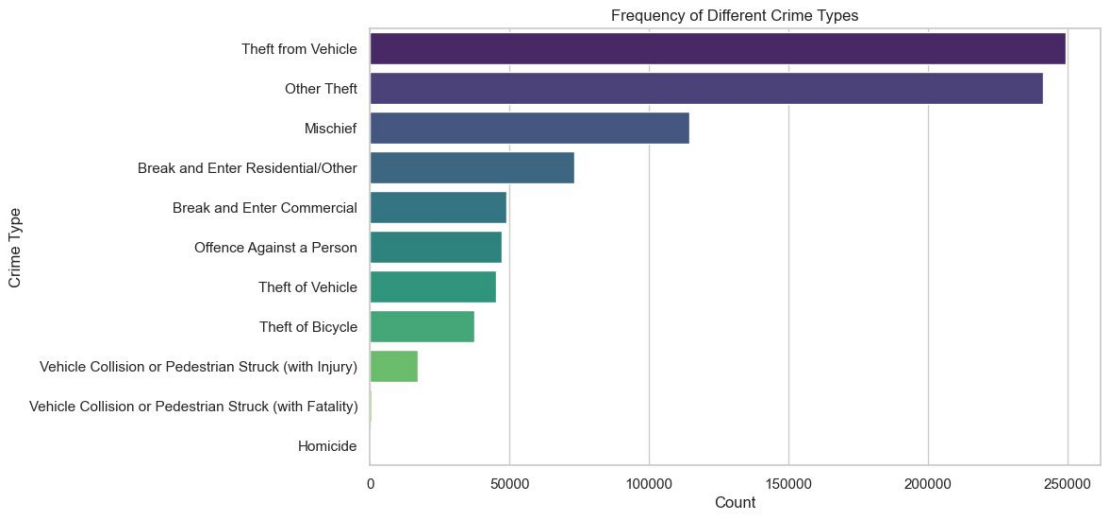
Strategic Analysis & Data Roadmap

Methodology

EDA

Urban Dynamics: Crime

- **The Datasets: Analyzed 875,000+ crime records.**
- **Crime Follows Commerce: Incidents are overwhelmingly property-related (Theft from Vehicle).**



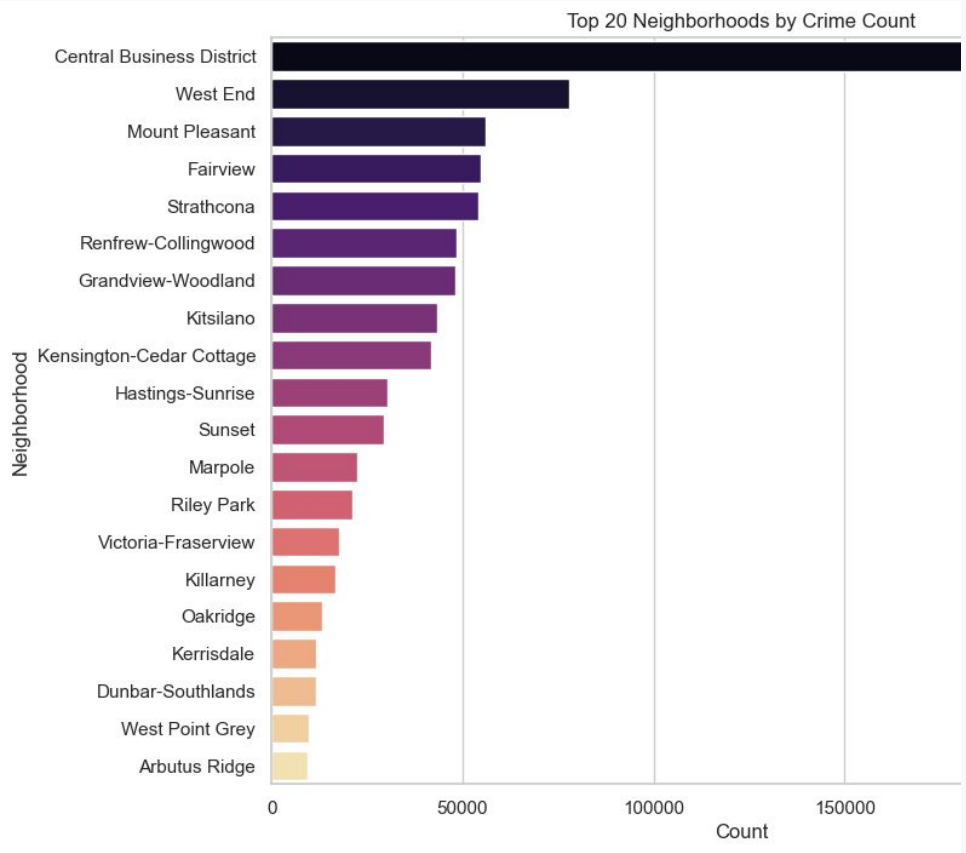
Strategic Analysis & Data Roadmap

Methodology

EDA

Urban Dynamics: Crime

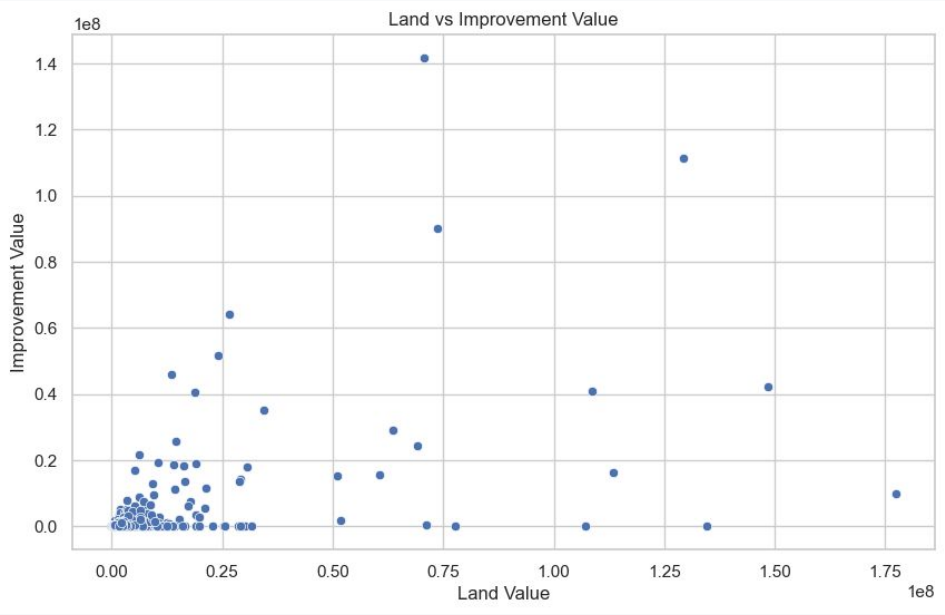
- **Crime Follows Commerce: These cluster heavily in the Central Business District due to high commercial foot traffic.**

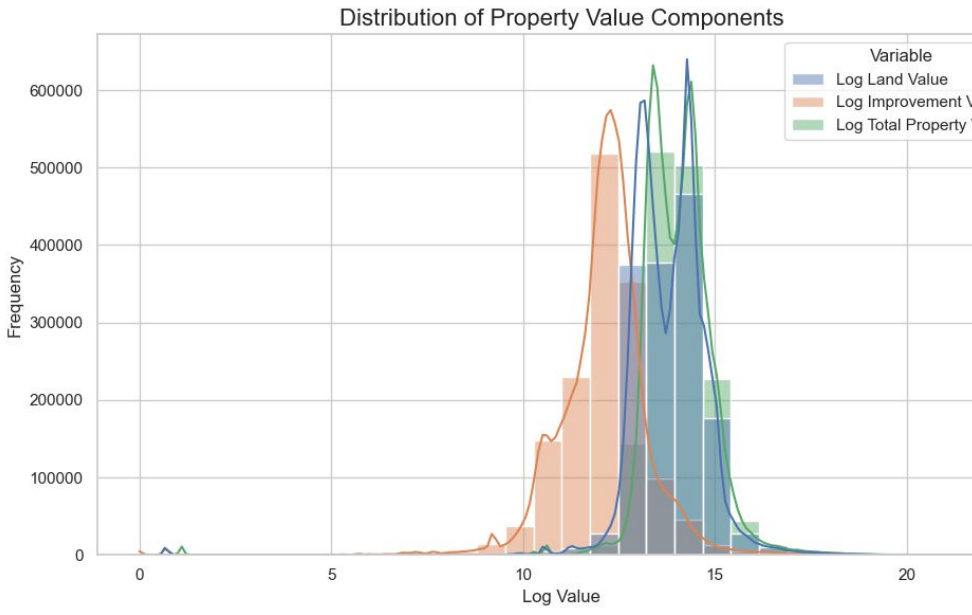


Strategic Analysis & Data Roadmap

Methodology

EDA



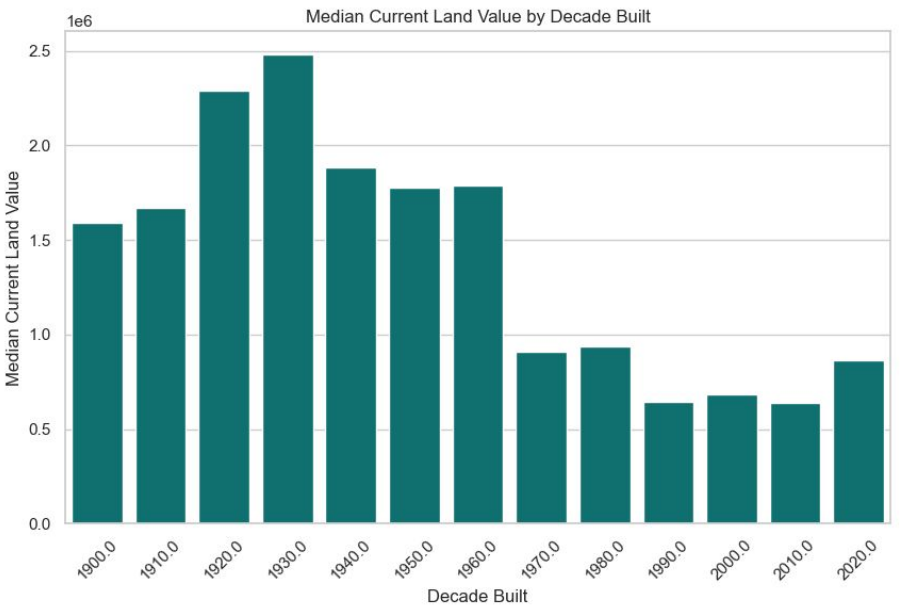


Strategic Analysis & Data Roadmap

Methodology
EDA

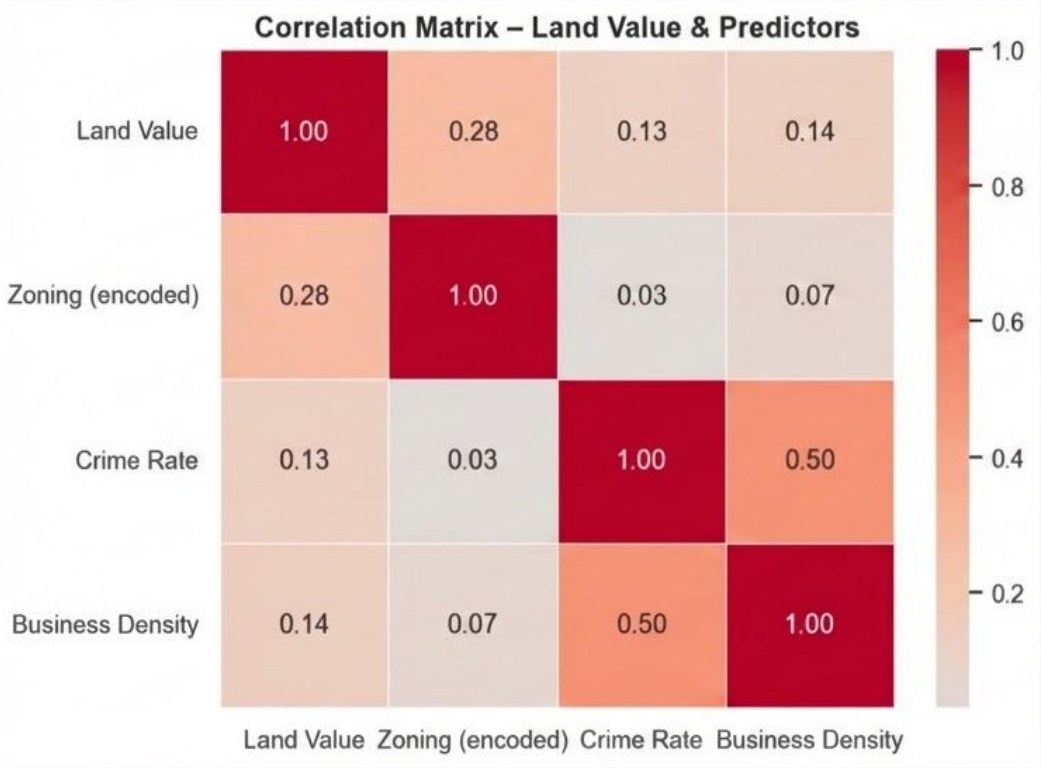
The Historical Valuation

- **Age vs. Value: Plotted Median Current Land Value against the Decade Built.**
- **The 1920s Peak: Properties built in the 1920s and 1930s command the highest median land values today (near $2.5M).**



Data Preparation & Modeling Approach

Methodology
Machine Learning



Goal: Using Multiple Linear Regression to predict the land value at Vancouver.

Features:

- Zoning classification
- Crime rate
- Business density

Data Processing: cutting off 1% of the minimum and 1% of maximum land values to eliminate excessively expensive properties and inflated land's prices.

Data Preparation & Modeling Approach

Methodology
Machine Learning

```
MODEL EVALUATION METRICS
R² Score: 0.0990
RMSE    : $1,309,987
MAE     : $793,910
```

```
Coefficient Table (standardised and raw):
          Feature   Coefficient  Raw Coefficient (per unit)
  zoning_encoded 375244.911269              119148.949436
      crime_rate 112390.229196                 172.475428
business_density 114773.879348                 859.401004
```

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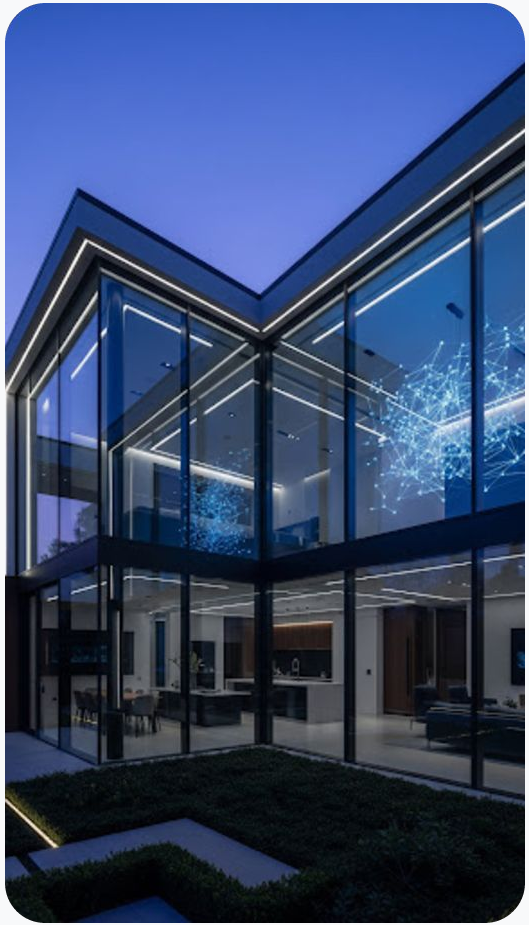


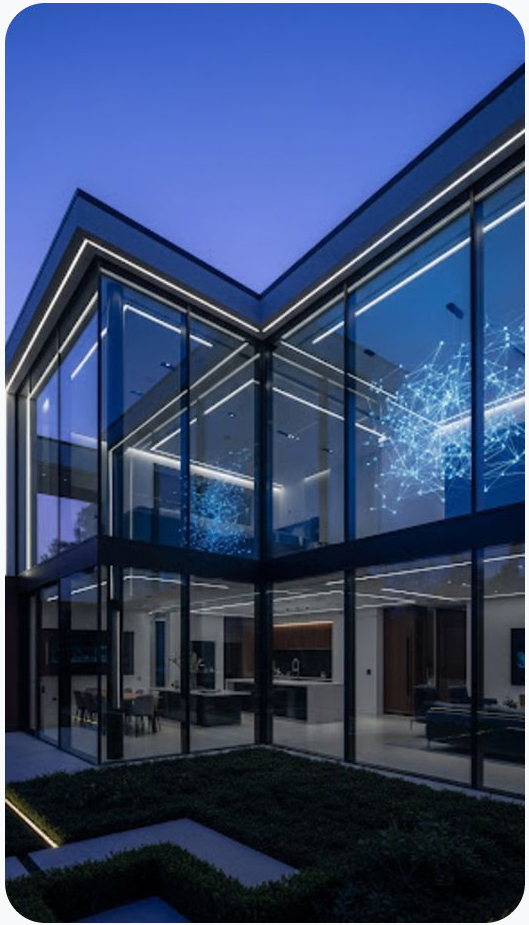
Strategic Analysis & Data Roadmap

Key Findings

Market Overview

Extreme Market Skew: The Vancouver market is heavily right-skewed. The overall median land value is $993,000,

Land Dominates Structure: At higher price tiers, land value completely eclipses improvement (building) value. You are paying for the dirt, not the house.



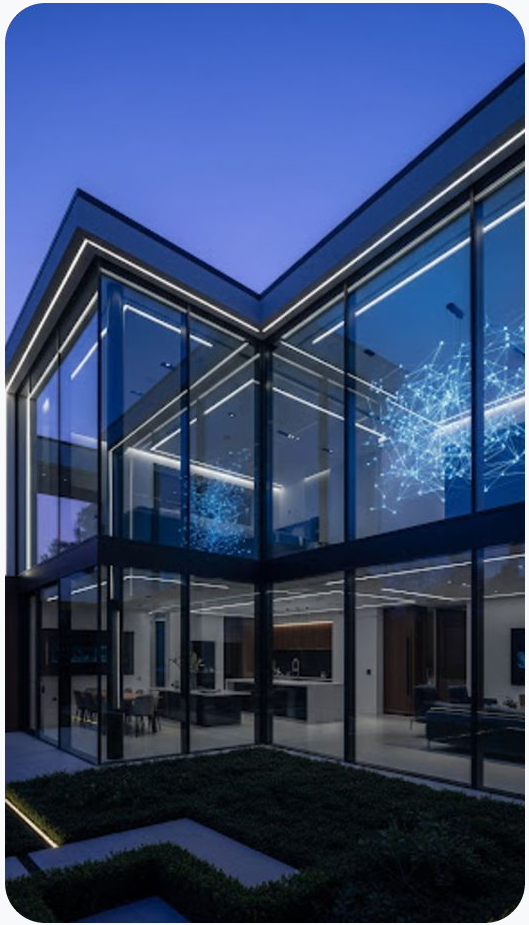
Strategic Analysis & Data Roadmap

Key Findings

Neighborhood Features

The Business Effect: There is a clear positive trend between business activity and land value.

The Crime Paradox: Higher crime rates do not strictly mean lower land values. The trendline actually shows a slight positive correlation.



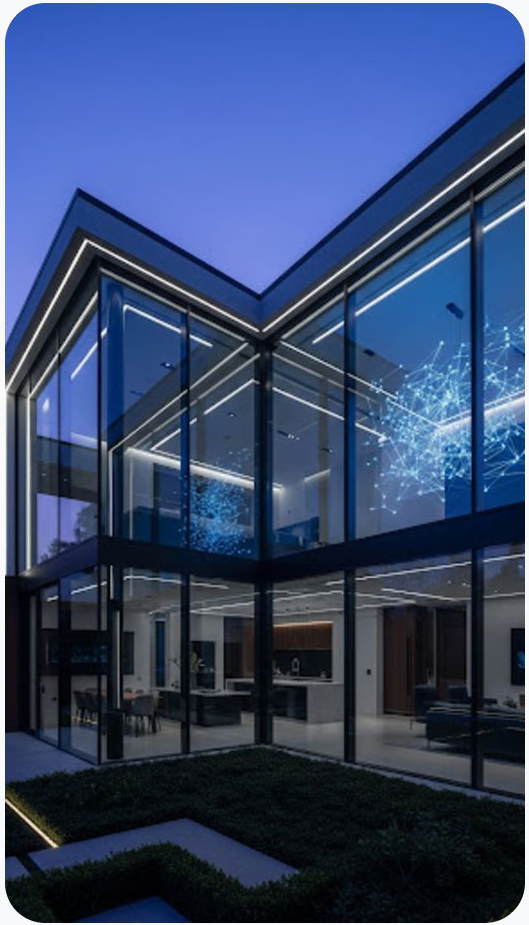
Strategic Analysis & Data Roadmap

Key Findings

Model Limitations

Multiple Linear Regression: Our model yielded an R^2 of 0.099. Features (zoning, crime, business density) only explain about 10% of a property's price variance.

High-End Market Variance: The model struggles most with luxury markets (properties over $3M). It is too volatile to predict with neighborhood-level data.



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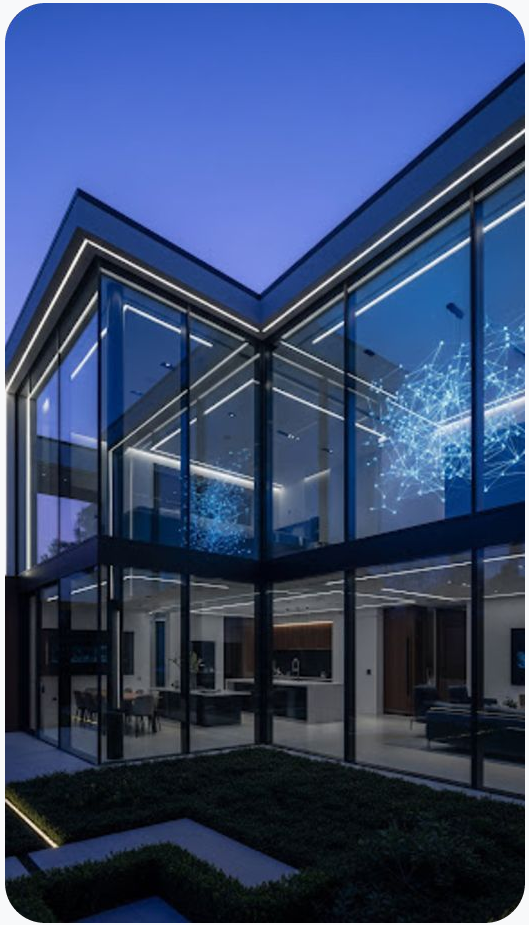


Strategic Analysis & Data Roadmap

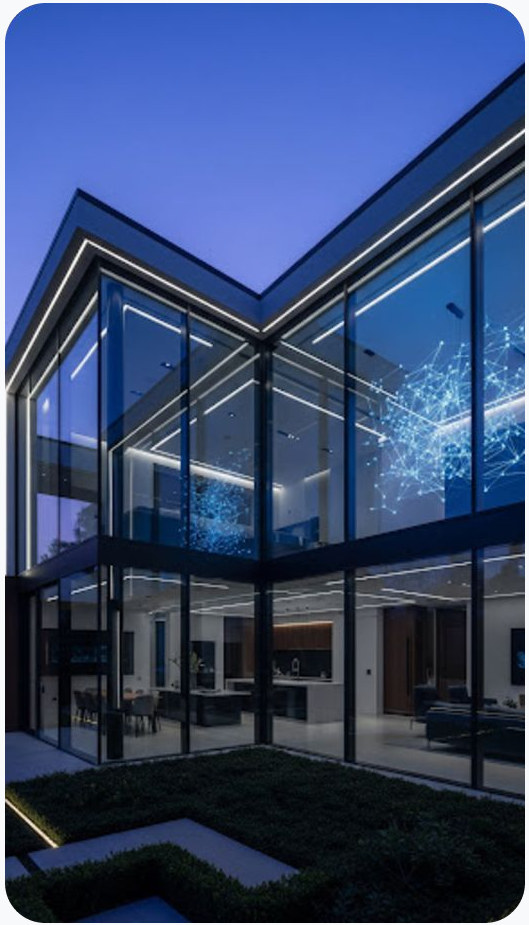
Business Recommendations

Strategic actions based on predictive demand and market trends

- **Location matters most**
  Focus on land value and high-demand areas for better returns

- **Target business-active zones**
  Higher economic activity → higher property value

- **Don’t rely only on crime**
  High-demand areas maintain value despite crime rates



Strategic Analysis & Data Roadmap

Business Recommendations

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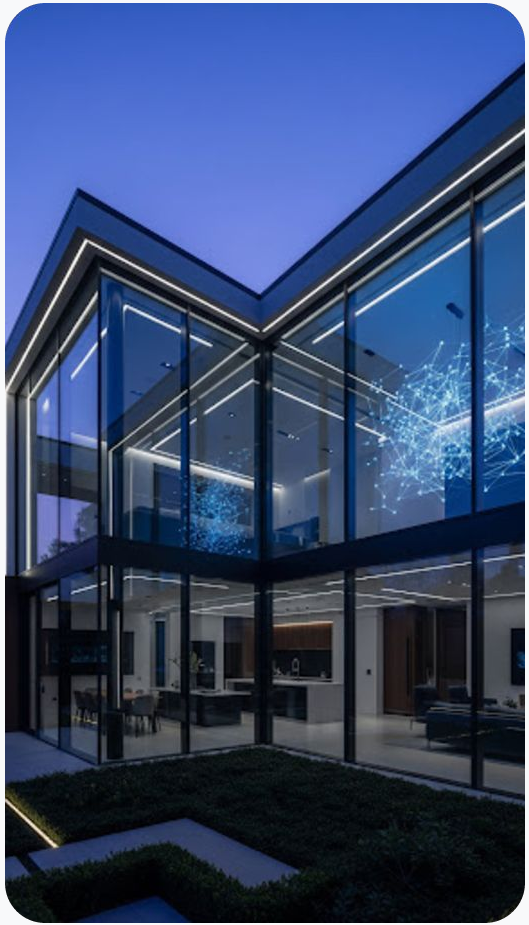
•Address affordability issues
Market skew suggests need for housing accessibility strategies

•Improve the model
Add features like zoning, amenities, and transport access



Thank you for listening!

Any questions?